

# ELITE IGCSE ACADEMY

*Student Past Paper Solutions*

## Jan 2021 P2HR Worked Solutions

Jan 2021 | P2HR

25 questions   ■   question image followed by worked solution

PREPARED BY

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**PAST PAPER QUESTION****Q#: 1****Marks: 8****TOPIC: Expanding Brackets**

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**1**  $w = 5y^2 - y^3$

(a) Work out the value of  $w$  when  $y = -2$ 

$w = \dots\dots\dots$   
(2)

(b) Factorise fully  $8p^2 - 2p$ 

$\dots\dots\dots$   
(2)

(c) Expand  $4t(3t - 2)$ 

$\dots\dots\dots$   
(2)

(d) Expand and simplify  $(5x - 2)(x + 4)$ 

$\dots\dots\dots$   
(2)

**(Total for Question 1 is 8 marks)**

**PAST PAPER SOLUTION****Q#: 1** **Marks: 8****TOPIC: Expanding Brackets**

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**WORKED SOLUTION***Dr. Eslam Ahmed***1. Expand brackets**For  $w = 5y^2 - y^3$  and  $y = -2$ :

$$w = 5(-2)^2 - (-2)^3 = 20 + 8 = 28$$

$$8p^2 - 2p = 2p(4p - 1)$$

$$4t(3t - 2) = 12t^2 - 8t$$

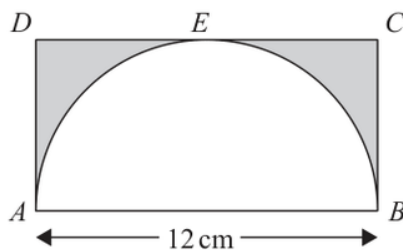
$$(5x - 2)(x + 4) = 5x^2 + 18x - 8$$

**FINAL ANSWER**28,  $2p(4p - 1)$ ,  $12t^2 - 8t$ ,  $5x^2 + 18x - 8$

**PAST PAPER QUESTION****Q#: 2** **Marks: 3****TOPIC: Area & Perimeter**

Jan 2021 P2HR - Jan 2021 | P2HR

- 2 The diagram shows a rectangle  $ABCD$  and a semicircle with diameter  $AB$  where  $AB = 12$  cm. The point  $E$  lies on  $DC$  and also on the semicircle.

Diagram **NOT**  
accurately drawn

Work out the area of the shaded region.  
Give your answer correct to 3 significant figures.

.....  $\text{cm}^2$ 

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**(Total for Question 2 is 3 marks)**

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**PAST PAPER SOLUTION****Q#: 2****Marks: 3****TOPIC: Area & Perimeter**

Jan 2021 P2HR - Jan 2021 | P2HR

**WORKED SOLUTION***Dr. Eslam Ahmed***1. Calculate area**

The semicircle has diameter 12 cm, so its radius is 6 cm.

**2. The rectangle has area**

The rectangle has area

$$12 \times 6 = 72$$

**3. Area of the semicircle**

Area of the semicircle:

$$\frac{1}{2}\pi(6)^2 = 18\pi$$

**4. Shaded area**

Shaded area:

$$72 - 18\pi = 15.451\dots$$

**FINAL ANSWER**

15.5 cm<sup>2</sup>.

**PAST PAPER QUESTION****Q#: 3****Marks: 3****TOPIC: Set Notation & Venn Diagrams**

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- 3**  $\mathcal{E} = \{21, 22, 23, 24, 25, 26, 27, 28, 29, 30\}$   
 $A = \{22, 24, 26, 28, 30\}$   
 $B = \{21, 24, 27, 30\}$

(a) List the members of the set

(i)  $A \cap B$

(ii)  $A'$

.....  
(2)

$C = \{23, 25, 29\}$

(b) Using set notation, find an expression for  $C$  in terms of  $A$  and  $B$ ......  
(1).....  
(Total for Question 3 is 3 marks)

**PAST PAPER SOLUTION****Q#: 3****Marks: 3****TOPIC: Set Notation & Venn Diagrams**

Jan 2021 P2HR - Jan 2021 | P2HR

**WORKED SOLUTION***Dr. Eslam Ahmed***1. Calculate value**

$$A \cap B = \{24, 30\}$$

**2. Set is**

The complement of  $A$ , using the given universal set, is

$$A' = \{21, 23, 25, 27, 29\}$$

**3. The set**

The set

$$C = \{23, 25, 29\}$$

**4. Calculate value**

is everything not in  $A$  and not in  $B$ , so

$$C = A' \cap B'$$

**5. Calculate value**

Equivalently,

$$C = (A \cup B)'$$

**FINAL ANSWER**

$A \cap B = \{24, 30\}$ ,  $A' = \{21, 23, 25, 27, 29\}$ , and  $C = A' \cap B'$ .

**PAST PAPER QUESTION****Q#: 4****Marks: 4**TOPIC: **Algebraic Roots & Indices**

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4 (a) Simplify  $(3k^2)^4$ .....  
(2)(b) Simplify  $(21m^4n) \div (3n^{-5})$ .....  
(2)

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**(Total for Question 4 is 4 marks)**

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EA

**PAST PAPER SOLUTION**

Q#: 4

Marks: 4

TOPIC: **Algebraic Roots & Indices**

Jan 2021 P2HR - Jan 2021 | P2HR

**WORKED SOLUTION**

Dr. Eslam Ahmed

**1. Use index laws**

For part (a),

$$\begin{aligned}(3k^2)^4 &= 3^4(k^2)^4 \\ &= 81k^8\end{aligned}$$

**2. Use index laws**

For part (b),

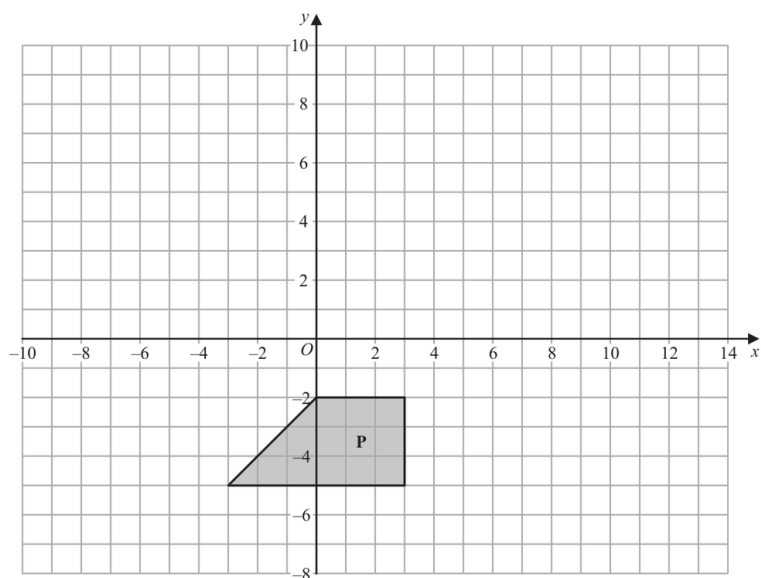
$$\begin{aligned}(21m^4n) \div (3n^{-5}) &= \frac{21m^4n}{3n^{-5}} \\ &= 7m^4n^{1-(-5)} \\ &= 7m^4n^6\end{aligned}$$

**FINAL ANSWER**(a)  $81k^8$ , (b)  $7m^4n^6$ .

**PAST PAPER QUESTION****Q#: 5** **Marks: 6****TOPIC: Transformations**

Jan 2021 P2HR - Jan 2021 | P2HR

- 5 Here is a shape **P** drawn on a grid of squares.



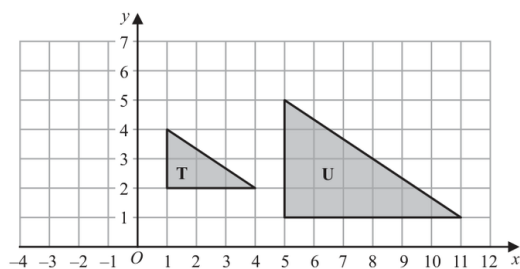
- (a) On the grid, rotate shape **P**  $180^\circ$  about the point  $(-3, 2)$ .  
Label the new shape **Q**.

(2)

- (b) On the grid, translate shape **P** by the vector  $\begin{pmatrix} 10 \\ 8 \end{pmatrix}$ .  
Label the new shape **R**.

(1)

Here are triangle **T** and triangle **U** drawn on a grid of squares.



- (c) Describe fully the single transformation that maps triangle **T** onto triangle **U**.

(3)

(Total for Question 5 is 6 marks)

**PAST PAPER SOLUTION****Q#: 5****Marks: 6****TOPIC: Transformations**

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**WORKED SOLUTION***Dr. Eslam Ahmed***1. (a) For a rotation about**(a) For a  $180^\circ$  rotation about  $(-3, 2)$ ,

$$(x, y) \mapsto (-6 - x, 4 - y)$$

**2. The image has vertices**

The image has vertices

$$(-3, 9), (-9, 9), (-9, 6), (-6, 6)$$

**3. (b) Translate by**

(b) Translate by

$$\begin{pmatrix} 10 \\ 8 \end{pmatrix}$$

**4. The image has vertices**

The image has vertices

$$(7, 3), (13, 3), (13, 6), (10, 6)$$

**5. Use trigonometry**(c) Triangle  $T$  is mapped to triangle  $U$  by an enlargement with scale factor 2 and centre  $(-3, 3)$ .**FINAL ANSWER**(a) vertices  $(-3, 9), (-9, 9), (-9, 6), (-6, 6)$ ; (b) vertices  $(7, 3), (13, 3), (13, 6), (10, 6)$ ; (c) enlargement, scale factor 2, centre  $(-3, 3)$ .

**PAST PAPER QUESTION****Q#: 6****Marks: 3****TOPIC: Percentages**

Jan 2021 P2HR - Jan 2021 | P2HR

- 6 On Wednesday, the price of 1 litre of petrol was £1.26  
The price of petrol on Wednesday was 5% more than the price of petrol on the previous Monday.

Calculate the price of 30 litres of petrol on the previous Monday.

£.....

(Total for Question 6 is 3 marks)

EA

**PAST PAPER SOLUTION****Q#: 6** **Marks: 3**TOPIC: **Percentages**

Jan 2021 P2HR - Jan 2021 | P2HR

**WORKED SOLUTION***Dr. Eslam Ahmed***1. Wednesday's price was more**

Wednesday's price was 5% more than Monday's price.

**2. Calculate percentage**

So

$$1.26 = 1.05 \times \text{Monday price}$$

$$\text{Monday price} = \frac{1.26}{1.05} = 1.20$$

**3. The cost of litres on**

The cost of 30 litres on Monday was

$$30 \times 1.20 = 36$$

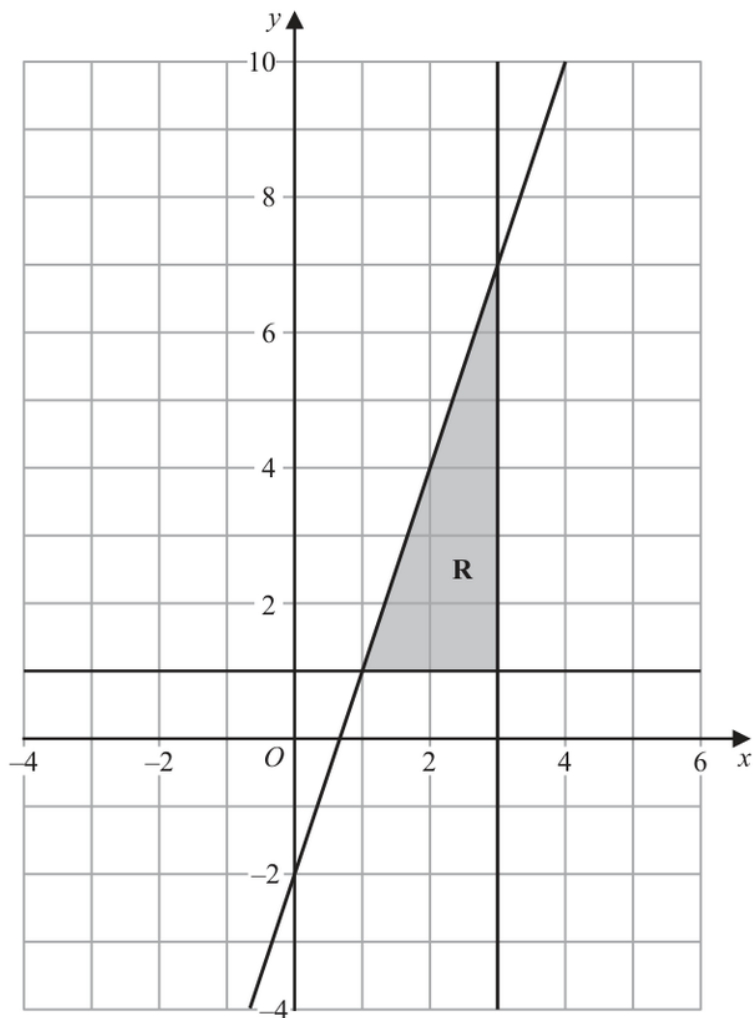
**FINAL ANSWER****£36**

**PAST PAPER QUESTION****Q#: 7** **Marks: 3****TOPIC: Graphing Inequalities**

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- 7 The shaded region **R**, shown in the diagram below, is bounded by the straight line with equation  $y = 3x - 2$  and by two other straight lines.

Write down the three inequalities that define region **R**.



(Total for Question 7 is 3 marks)

**PAST PAPER SOLUTION****Q#: 7** **Marks: 3****TOPIC: Graphing Inequalities**

Jan 2021 P2HR - Jan 2021 | P2HR

**WORKED SOLUTION***Dr. Eslam Ahmed***1. The sloping boundary is**

The sloping boundary is

$$y = 3x - 2$$

**2. The shaded region is below**

The shaded region is below this line:

$$y \leq 3x - 2$$

**3. The vertical boundary is so**

The vertical boundary is

$$x = 3$$

so the region has

$$x \leq 3$$

**4. The lower boundary is the**The lower boundary is the  $x$ -axis, so

$$y \geq 0$$

**FINAL ANSWER**

$$y \leq 3x - 2, x \leq 3, y \geq 0.$$

**PAST PAPER QUESTION****Q#: 8** **Marks: 3****TOPIC: Powers, Roots & Standard Form**

Jan 2021 P2HR - Jan 2021 | P2HR

- 8 The table gives the length of the coastline, in kilometres, of each of five oceans.

| Ocean    | Length of coastline (km) |
|----------|--------------------------|
| Arctic   | $4.539 \times 10^4$      |
| Atlantic | $1.119 \times 10^5$      |
| Pacific  | $1.357 \times 10^5$      |
| Indian   | $6.653 \times 10^4$      |
| Southern | $1.797 \times 10^4$      |

- (a) Which ocean has the greatest length of coastline?

.....  
(1)

- (b) Calculate the difference between the length of the Atlantic Ocean's coastline and the length of the Southern Ocean's coastline.  
Give your answer in standard form.

..... km  
(2)

(Total for Question 8 is 3 marks)

**PAST PAPER SOLUTION****Q#: 8** **Marks: 3****TOPIC: Powers, Roots & Standard Form**

Jan 2021 P2HR - Jan 2021 | P2HR

**WORKED SOLUTION***Dr. Eslam Ahmed***1. The greatest coastline is the**

The greatest coastline is the largest number in standard form.

$$1.357 \times 10^5$$

**2. Convert standard form**

is the largest, so the answer is the Pacific Ocean.

**3. Convert standard form**

For part (b),

$$1.119 \times 10^5 = 111900$$

$$1.797 \times 10^4 = 17970$$

**4. Difference**

Difference:

$$111900 - 17970 = 93930$$

$$93930 = 9.393 \times 10^4$$

**FINAL ANSWER**

(a) Pacific Ocean, (b)  $9.393 \times 10^4$  km

**PAST PAPER QUESTION****Q#: 9****Marks: 3****TOPIC: Solving Quadratic Equations**

Jan 2021 P2HR - Jan 2021 | P2HR

- 9 Solve  $x^2 - 21x + 20 = 0$   
Show your working clearly.

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(Total for Question 9 is 3 marks)

EA

**PAST PAPER SOLUTION****Q#: 9****Marks: 3****TOPIC: Solving Quadratic Equations**

Jan 2021 P2HR - Jan 2021 | P2HR

**WORKED SOLUTION***Dr. Eslam Ahmed***1. Solve quadratic equation**

$$x^2 - 21x + 20 = 0$$

**2. Factorise**

Factorise:

$$(x - 20)(x - 1) = 0$$

**3. Solve quadratic equation**

So

$$x = 20 \quad \text{or} \quad x = 1$$

**FINAL ANSWER**

$$x = 20 \text{ or } x = 1.$$

**PAST PAPER QUESTION****Q#: 10****Marks: 4****TOPIC: Statistics Toolkit**

Jan 2021 P2HR - Jan 2021 | P2HR

- 10** A mathematics teacher at a school asked a group of students how far, in kilometres, each student had travelled to get to school that day.

The table gives information about their answers.

| Distance travelled ( $d$ km) | Number of students |
|------------------------------|--------------------|
| $0 < d \leq 2$               | $x$                |
| $2 < d \leq 4$               | 11                 |
| $4 < d \leq 6$               | 8                  |
| $6 < d \leq 8$               | 6                  |
| $8 < d \leq 10$              | 5                  |

The teacher calculated that an estimate for the mean distance travelled by the whole group of students was 4.25 km.

Work out the value of  $x$ .

Show your working clearly.

$x = \dots\dots\dots$

**(Total for Question 10 is 4 marks)**

**PAST PAPER SOLUTION****Q#: 10****Marks: 4****TOPIC: Statistics Toolkit**

Jan 2021 P2HR - Jan 2021 | P2HR

**WORKED SOLUTION***Dr. Eslam Ahmed***1. Use class midpoints**

Use class midpoints 1, 3, 5, 7, 9.

**2. The total frequency is**

The total frequency is

$$x + 11 + 8 + 6 + 5 = x + 30$$

**3. The total estimated distance is**

The total estimated distance is

$$1x + 3(11) + 5(8) + 7(6) + 9(5) = x + 160$$

**4. The estimated mean is**

The estimated mean is 4.25:

$$\frac{x + 160}{x + 30} = 4.25$$

$$x + 160 = 4.25x + 127.5$$

$$32.5 = 3.25x$$

$$x = 10$$

**FINAL ANSWER** $x = 10.$

**PAST PAPER QUESTION****Q#: 11****Marks: 4****TOPIC: Circles, Arcs & Sectors**

Jan 2021 P2HR - Jan 2021 | P2HR

- 11** A circle centre  $O$  has radius 9 cm.

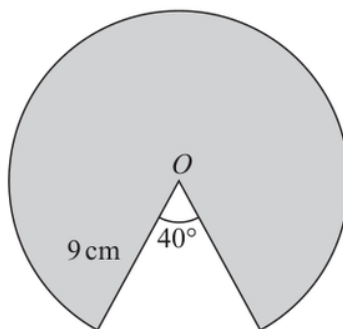


Diagram **NOT**  
accurately drawn

Calculate the perimeter of the shaded sector of the circle.  
Give your answer correct to 3 significant figures.

..... cm

(Total for Question 11 is 4 marks)

**PAST PAPER SOLUTION****Q#: 11****Marks: 4****TOPIC: Circles, Arcs & Sectors**

Jan 2021 P2HR - Jan 2021 | P2HR

**WORKED SOLUTION***Dr. Eslam Ahmed***1. The shaded sector is the**

The shaded sector is the major sector:

$$360^\circ - 40^\circ = 320^\circ$$

**2. Arc length**

Arc length:

$$\frac{320}{360} \times 2\pi(9) = 16\pi$$

**3. Perimeter**

Perimeter:

$$16\pi + 2(9) = 68.265\dots$$

**FINAL ANSWER**

68.3 cm.

**PAST PAPER QUESTION****Q#: 12****Marks: 4****TOPIC: Simultaneous Equations**

Jan 2021 P2HR - Jan 2021 | P2HR

- 12** Solve the simultaneous equations  $2x + 7y = 17$   
 $5x + 3y = -1$

Show clear algebraic working.

 $x = \dots\dots\dots$  $y = \dots\dots\dots$ 

**(Total for Question 12 is 4 marks)**

EA

**PAST PAPER SOLUTION****Q#: 12****Marks: 4****TOPIC: Simultaneous Equations**

Jan 2021 P2HR - Jan 2021 | P2HR

**WORKED SOLUTION***Dr. Eslam Ahmed***1. Solve simultaneous equations**

$$2x + 7y = 17$$

$$5x + 3y = -1$$

**2. Multiply the first equation by**

Multiply the first equation by 5:

$$10x + 35y = 85$$

**3. Multiply the second equation by**

Multiply the second equation by 2:

$$10x + 6y = -2$$

**4. Subtract**

Subtract:

$$29y = 87$$

$$y = 3$$

**5. Solve simultaneous equations**Substitute into  $2x + 7y = 17$ :

$$2x + 21 = 17$$

$$2x = -4$$

$$x = -2$$

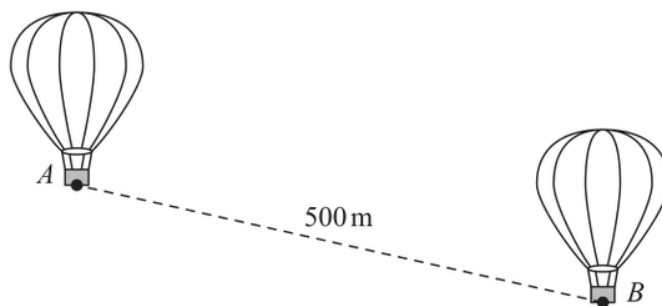
**FINAL ANSWER**

$$x = -2, y = 3.$$

**PAST PAPER QUESTION****Q#: 13** **Marks: 3****TOPIC: Right-Angled Triangles - Pythagoras & Trigonometry**

Jan 2021 P2HR - Jan 2021 | P2HR

- 13** The diagram shows two hot air balloons.  
 $A$  is a point on the base of one of the balloons and  $B$  is a point on the base of the other balloon.

Diagram **NOT**  
accurately drawn

The distance between  $A$  and  $B$  is 500 metres.  
The angle of depression of  $B$  from  $A$  is  $23^\circ$

Calculate the vertical height of  $A$  above  $B$ .  
Give your answer correct to one decimal place.

..... metres

**(Total for Question 13 is 3 marks)**

**PAST PAPER SOLUTION****Q#: 13****Marks: 3****TOPIC: Right-Angled Triangles - Pythagoras & Trigonometry**

Jan 2021 P2HR - Jan 2021 | P2HR

**WORKED SOLUTION***Dr. Eslam Ahmed***1. The vertical height is opposite**

The vertical height is opposite the  $23^\circ$  angle, and  $AB = 500$  m is the hypotenuse.

$$\sin 23^\circ = \frac{h}{500}$$

$$h = 500 \sin 23^\circ = 195.365 \dots$$

**FINAL ANSWER**

195.4 m.

EA

**PAST PAPER QUESTION****Q#: 14** **Marks: 4****TOPIC: Compound Interest & Depreciation**

Jan 2021 P2HR - Jan 2021 | P2HR

**14** Simon bought a house at the beginning of 2018

The value of Simon's house had decreased by 15% by the end of 2018

The house increased in value during both 2019 and 2020

The percentage increases in the value of the house during 2019 and 2020 were the same.

The value of Simon's house at the end of 2020 was 2.85% greater than the amount he paid for his house at the beginning of 2018

Calculate the percentage increase in the value of the house during 2019

.....%

**(Total for Question 14 is 4 marks)**

EA

**PAST PAPER SOLUTION****Q#: 14****Marks: 4****TOPIC: Compound Interest & Depreciation**

Jan 2021 P2HR - Jan 2021 | P2HR

**WORKED SOLUTION***Dr. Eslam Ahmed***1. Let the original house value**Let the original house value be  $V$ .**2. Use compound interest**

After the 15% decrease in 2018:

$$0.85V$$

**3. Use compound interest**Let the percentage increase in each of 2019 and 2020 be  $r$ .**4. The final value is greater**

The final value is 2.85% greater than the original value:

$$1.0285V$$

**5. Use compound interest**

So

$$0.85V(1+r)^2 = 1.0285V$$

$$(1+r)^2 = \frac{1.0285}{0.85} = 1.21$$

$$1+r = \sqrt{1.21} = 1.1$$

$$r = 0.1 = 10\%$$

**FINAL ANSWER**

10%

**PAST PAPER QUESTION****Q#: 15****Marks: 4****TOPIC: Algebraic Proof**

Jan 2021 P2HR - Jan 2021 | P2HR

**15** Prove algebraically that the product of any two odd numbers is always an odd number.

(Total for Question 15 is 4 marks)

EA

**PAST PAPER SOLUTION**

Q#: 15 Marks: 4

TOPIC: **Algebraic Proof**

Jan 2021 P2HR - Jan 2021 | P2HR

**WORKED SOLUTION**

Dr. Eslam Ahmed

**1. Let the two odd numbers**

Let the two odd numbers be

$$2m + 1 \quad \text{and} \quad 2n + 1$$

**2. Their product is**

Their product is

$$\begin{aligned} & (2m + 1)(2n + 1) \\ &= 4mn + 2m + 2n + 1 \\ &= 2(2mn + m + n) + 1 \end{aligned}$$

**3. This is of the form**

This is of the form

$$2k + 1$$

where  $k$  is an integer.**4. Use algebraic proof asks general**

So the product is odd.

**FINAL ANSWER**

The product of any two odd numbers is always odd.

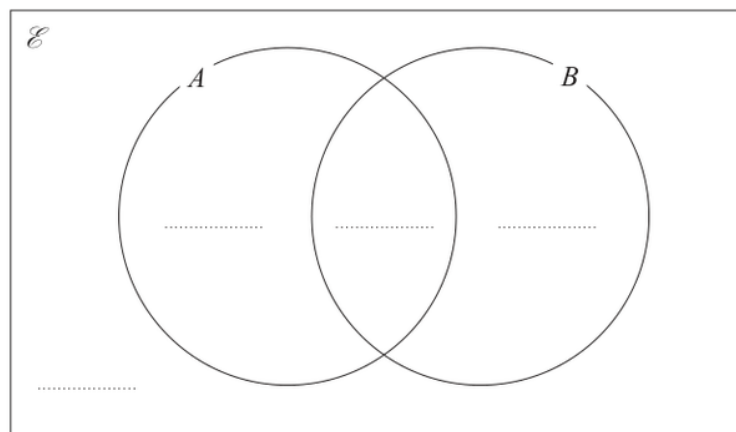
**PAST PAPER QUESTION****Q#: 16****Marks: 5****TOPIC: Probability Diagrams - Venn & Tree Diagrams**

Jan 2021 P2HR - Jan 2021 | P2HR

**16** Two events  $A$  and  $B$  are such that  $n(A) = 62$   $n(B) = 30$  and  $n(A \cup B) = 68$

Given that  $n(\mathcal{E}) = 80$

(a) complete the Venn diagram to show the number of elements in each region.



(2)

An element is chosen at random from  $\mathcal{E}$ .

(b) Using the Venn diagram, find the probability that this element is in

(i)  $A \cap B$

.....  
(1)

(ii)  $A \cup B'$

.....  
(2)

**(Total for Question 16 is 5 marks)**

**PAST PAPER SOLUTION****Q#: 16****Marks: 5****TOPIC: Probability Diagrams - Venn & Tree Diagrams**

Jan 2021 P2HR - Jan 2021 | P2HR

**WORKED SOLUTION***Dr. Eslam Ahmed***1. Calculate probability**

$$n(A \cap B) = 62 + 30 - 68 = 24$$

**2. Calculate probability**

So the Venn diagram has:

$$A \text{ only} = 38, \quad B \text{ only} = 6, \quad A \cap B = 24, \quad \text{outside} = 12$$

$$P(A \cap B) = \frac{24}{80} = \frac{3}{10}$$

$$P(A \cup B') = \frac{74}{80} = \frac{37}{40}$$

**FINAL ANSWER**

$$\frac{3}{10}, \frac{37}{40}$$

**PAST PAPER QUESTION****Q#: 17** **Marks: 9****TOPIC: Functions**

Jan 2021 P2HR - Jan 2021 | P2HR

**17** The functions  $f$  and  $g$  are defined as

$$f(x) = x^2 + 6$$

$$g(x) = x - 10$$

(a) Find  $fg(3)$ .....  
(2)(b) Solve the equation  $fg(x) = f(x)$   
Show clear algebraic working......  
(3)The function  $h$  is defined as

$$h(x) = \frac{2x - 4}{x}$$

(c) State the value of  $x$  that cannot be included in the domain of  $h$ .....  
(1)(d) Express the inverse function  $h^{-1}$  in the form  $h^{-1}(x) = \dots$ 

$$h^{-1}(x) = \dots$$

(3)

**(Total for Question 17 is 9 marks)**

**PAST PAPER SOLUTION****Q#: 17****Marks: 9****TOPIC: Functions**

Jan 2021 P2HR - Jan 2021 | P2HR

**WORKED SOLUTION***Dr. Eslam Ahmed***1. Solve quadratic equation**

$$f(x) = x^2 + 6, \quad g(x) = x - 10$$

**2. (a)**

(a)

$$g(3) = 3 - 10 = -7$$

$$fg(3) = f(-7) = (-7)^2 + 6 = 55$$

**3. (b)**

(b)

$$fg(x) = f(g(x)) = (x - 10)^2 + 6$$

**4. Solve**Solve  $fg(x) = f(x)$ :

$$(x - 10)^2 + 6 = x^2 + 6$$

$$x^2 - 20x + 100 = x^2$$

$$-20x + 100 = 0$$

$$x = 5$$

**5. (c)**

(c)

$$h(x) = \frac{2x - 4}{x}$$

**6. Solve quadratic equation**The denominator cannot be zero, so  $x = 0$  cannot be included.**7. (d) Let**

(d) Let

$$y = \frac{2x - 4}{x}$$

$$y = 2 - \frac{4}{x}$$

$$y - 2 = -\frac{4}{x}$$

$$x = \frac{4}{2 - y}$$

### 8. Solve quadratic equation

Therefore

$$h^{-1}(x) = \frac{4}{2 - x}$$

#### FINAL ANSWER

$fg(3) = 55$ ,  $x = 5$ , excluded value 0, and  $h^{-1}(x) = \frac{4}{2 - x}$ .

EA

**PAST PAPER QUESTION****Q#: 18****Marks: 5****TOPIC: Algebraic Fractions**

Jan 2021 P2HR - Jan 2021 | P2HR

**18** Solve the equation

$$\frac{5}{x+2} + \frac{3}{x^2+2x} = 2$$

Show clear algebraic working.

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**(Total for Question 18 is 5 marks)**

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EA

**PAST PAPER SOLUTION****Q#: 18** **Marks: 5**TOPIC: **Algebraic Fractions**

Jan 2021 P2HR - Jan 2021 | P2HR

**WORKED SOLUTION***Dr. Eslam Ahmed***1. Solve equation**

$$\frac{5}{x+2} + \frac{3}{x^2+2x} = 2$$

**2. Solve equation**

Since

$$x^2 + 2x = x(x + 2)$$

**3. multiply by**multiply by  $x(x + 2)$ :

$$5x + 3 = 2x(x + 2)$$

$$5x + 3 = 2x^2 + 4x$$

$$2x^2 - x - 3 = 0$$

**4. Factorise**

Factorise:

$$(2x - 3)(x + 1) = 0$$

$$x = \frac{3}{2} \quad \text{or} \quad x = -1$$

**5. Both values are allowed because**Both values are allowed because  $x \neq 0$  and  $x \neq -2$ .**FINAL ANSWER**

$$x = \frac{3}{2} \text{ or } x = -1.$$

**PAST PAPER QUESTION****Q#: 19****Marks: 7****TOPIC: Algebraic Roots & Indices**

Jan 2021 P2HR - Jan 2021 | P2HR

**19 (a)** Simplify  $8^2 \times \sqrt[3]{4^6}$ 

Give your answer in the form  $2^a$  where  $a$  is an integer.  
 Show each stage of your working clearly.

.....  
(3)

Given that  $n^{\left(-\frac{4}{5}\right)} = \left(\frac{1}{2}\right)^4$  where  $n > 0$

(b) find the value of  $n$ .

$n =$  .....  
 (4)

(Total for Question 19 is 7 marks)

**PAST PAPER SOLUTION****Q#: 19****Marks: 7****TOPIC: Algebraic Roots & Indices**

Jan 2021 P2HR - Jan 2021 | P2HR

**WORKED SOLUTION***Dr. Eslam Ahmed***1. Simplify surd**

$$8^2 = (2^3)^2 = 2^6$$

$$\sqrt[3]{4^6} = \sqrt[3]{(2^2)^6} = \sqrt[3]{2^{12}} = 2^4$$

**2. Simplify surd**

So

$$8^2 \times \sqrt[3]{4^6} = 2^6 \times 2^4 = 2^{10}$$

**3. Use index laws**

For part (b),

$$n^{-4/5} = \left(\frac{1}{2}\right)^4 = 2^{-4}$$

**4. Use index laws**

So

$$n^{4/5} = 2^4$$

**5. Raise both sides to the**Raise both sides to the power  $\frac{5}{4}$ :

$$n = (2^4)^{5/4} = 2^5 = 32$$

**FINAL ANSWER**(a)  $2^{10}$ , (b)  $n = 32$ .

**PAST PAPER QUESTION****Q#: 20****Marks: 5****TOPIC: Circles, Arcs & Sectors**

Jan 2021 P2HR - Jan 2021 | P2HR

**20**  $A$ ,  $B$  and  $C$  are points on a circle with centre  $O$ .

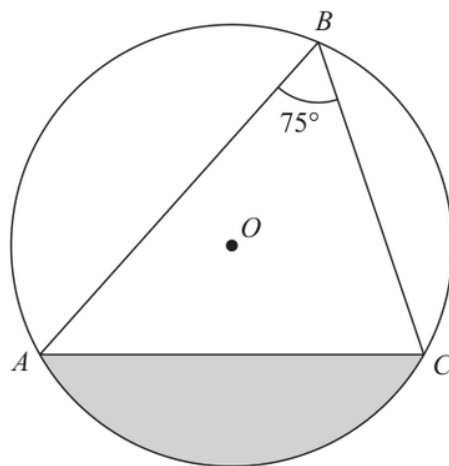


Diagram **NOT**  
accurately drawn

Angle  $ABC = 75^\circ$

The area of the shaded segment is  $200 \text{ cm}^2$

Calculate the radius of the circle.

Give your answer correct to 3 significant figures.

..... cm

(Total for Question 20 is 5 marks)

**PAST PAPER SOLUTION****Q#: 20****Marks: 5****TOPIC: Circles, Arcs & Sectors**

Jan 2021 P2HR - Jan 2021 | P2HR

**WORKED SOLUTION***Dr. Eslam Ahmed***1. Use trigonometry**

$$\angle ABC = 75^\circ$$

**2. The angle at the centre**

The angle at the centre is twice the angle at the circumference, so

$$\angle AOC = 2(75^\circ) = 150^\circ$$

**3. The shaded segment is**

The shaded segment is:

$$\text{sector } AOC - \text{triangle } AOC$$

**4. Let the radius be**

Let the radius be  $r$ .

**5. Sector area**

Sector area:

$$\frac{150}{360}\pi r^2$$

**6. Triangle area**

Triangle area:

$$\frac{1}{2}r^2 \sin 150^\circ$$

**7. Use trigonometry**

So

$$\frac{150}{360}\pi r^2 - \frac{1}{2}r^2 \sin 150^\circ = 200$$

$$r^2 \left( \frac{5\pi}{12} - \frac{1}{4} \right) = 200$$

$$r^2 = \frac{200}{\frac{5\pi}{12} - \frac{1}{4}}$$

$$r = 13.7426 \dots$$

**FINAL ANSWER**

13.7 cm

**PAST PAPER QUESTION****Q#: 20** **Marks: 5****TOPIC: Circles, Arcs & Sectors**

Jan 2021 P2HR - Jan 2021 | P2HR

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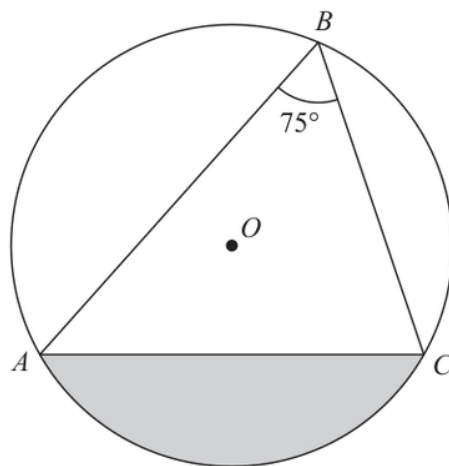


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Give your answer correct to 3 significant figures.

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Jan 2021 P2HR - Jan 2021 | P2HR

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$$r^2 = \frac{200}{\frac{5\pi}{12} - \frac{1}{4}}$$

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**FINAL ANSWER**

13.7 cm

**PAST PAPER QUESTION****Q#: 21** **Marks: 6****TOPIC: Probability Toolkit**

Jan 2021 P2HR - Jan 2021 | P2HR

- 21** A bag contains  $n$  beads.  
6 of the beads are red and the rest are blue.

Ravi is going to take at random 2 beads from the bag.

The probability that the 2 beads will be of the same colour is  $\frac{9}{17}$

Using algebra, and showing each stage of your working, calculate the value of  $n$ .

$n =$  .....

(Total for Question 21 is 6 marks)

**PAST PAPER SOLUTION****Q#: 21****Marks: 6****TOPIC: Probability Toolkit**

Jan 2021 P2HR - Jan 2021 | P2HR

**WORKED SOLUTION***Dr. Eslam Ahmed***1. There are beads altogether**There are  $n$  beads altogether.**2. Red beads**

Red beads:

$$6$$

**3. Blue beads**

Blue beads:

$$n - 6$$

**4. The probability of taking two**

The probability of taking two beads of the same colour is

$$P(RR) + P(BB) = \frac{9}{17}$$

**5. Use trigonometry**

So

$$\frac{6}{n} \cdot \frac{5}{n-1} + \frac{n-6}{n} \cdot \frac{n-7}{n-1} = \frac{9}{17}$$

$$\frac{30 + (n-6)(n-7)}{n(n-1)} = \frac{9}{17}$$

**6. Expand**

Expand:

$$(n-6)(n-7) = n^2 - 13n + 42$$

**7. Use trigonometry**

So

$$\frac{n^2 - 13n + 72}{n(n-1)} = \frac{9}{17}$$

**8. Cross multiply**

Cross multiply:

$$17(n^2 - 13n + 72) = 9n(n-1)$$

$$17n^2 - 221n + 1224 = 9n^2 - 9n$$

$$8n^2 - 212n + 1224 = 0$$

**9. Divide by 4**

Divide by 4:

$$2n^2 - 53n + 306 = 0$$

**10. Factor**

Factor:

$$(2n - 17)(n - 18) = 0$$

**11. Use trigonometry**

So

$$n = \frac{17}{2} \quad \text{or} \quad n = 18$$

**12. The number of beads must**

The number of beads must be a whole number, so

$$n = 18$$

**FINAL ANSWER**

|    |
|----|
| 18 |
|----|

**PAST PAPER QUESTION****Q#: 21****Marks: 6****TOPIC: Probability Toolkit**

Jan 2021 P2HR - Jan 2021 | P2HR

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$n =$  .....

(Total for Question 21 is 6 marks)

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Jan 2021 P2HR - Jan 2021 | P2HR

**WORKED SOLUTION***Dr. Eslam Ahmed***1. There are beads altogether**There are  $n$  beads altogether.**2. Red beads**

Red beads:

$$6$$

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Blue beads:

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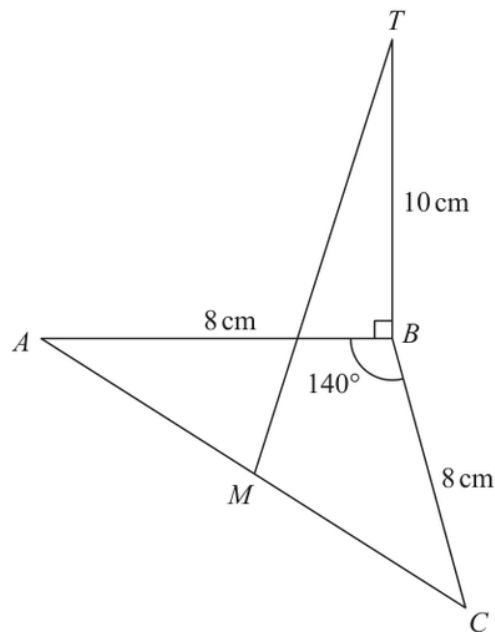
### FINAL ANSWER

18

**PAST PAPER QUESTION****Q#: 22****Marks: 4****TOPIC: 3D Pythagoras & Trigonometry**

Jan 2021 P2HR - Jan 2021 | P2HR

- 22**  $ABC$  is an isosceles triangle in a horizontal plane.  
The point  $T$  is vertically above  $B$ .

Diagram **NOT**  
accurately drawn

Angle  $ABC = 140^\circ$   
 $AB = BC = 8 \text{ cm}$   
 $TB = 10 \text{ cm}$   
 $M$  is the midpoint of  $AC$ .

Calculate the size of the angle between  $MT$  and the horizontal plane  $ABC$ .  
 Give your answer correct to one decimal place.

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(Total for Question 22 is 4 marks)

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**PAST PAPER SOLUTION****Q#: 22****Marks: 4****TOPIC: 3D Pythagoras & Trigonometry**

Jan 2021 P2HR - Jan 2021 | P2HR

**WORKED SOLUTION***Dr. Eslam Ahmed***1. Use trigonometry**

Since  $AB = BC = 8$ , triangle  $ABC$  is isosceles.

**2. The line from to the**

The line from  $B$  to the midpoint  $M$  bisects angle  $ABC$ .

$$\angle ABM = \frac{140^\circ}{2} = 70^\circ$$

**3. Use trigonometry**

In right triangle  $ABM$ ,

$$\cos 70^\circ = \frac{BM}{8}$$

$$BM = 8 \cos 70^\circ$$

**4. Use trigonometry**

Since  $T$  is vertically above  $B$ , the projection of  $MT$  onto the horizontal plane is  $MB$ .

**5. Let the required angle be**

Let the required angle be  $\theta$ .

$$\tan \theta = \frac{TB}{BM}$$

$$\tan \theta = \frac{10}{8 \cos 70^\circ}$$

$$\theta = 74.7^\circ$$

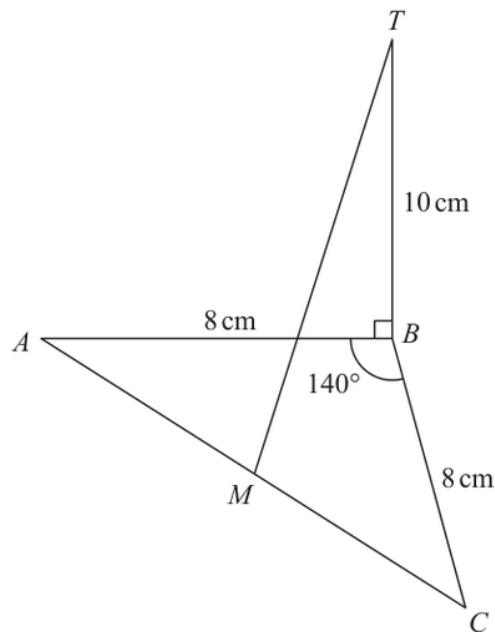
**FINAL ANSWER**

74.7°

**PAST PAPER QUESTION****Q#: 22****Marks: 4****TOPIC: 3D Pythagoras & Trigonometry**

Jan 2021 P2HR - Jan 2021 | P2HR

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 Give your answer correct to one decimal place.

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(Total for Question 22 is 4 marks)

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Jan 2021 P2HR - Jan 2021 | P2HR

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**FINAL ANSWER**

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